MIL-I-60279A(MU)
15 April 1969
SUPERSEDING
MIL-I-60279(MU)
21 June 1966

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED: DELAY, M4A1 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated: Delay, M4A1, Assembly, loaded with Cartridge, Delay, M46, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167	- Ammunition Data Card
MS-28741	- Hose Assembly - Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

F8595816	- Initiator, Cartridge Actuated: Delay, M4A1, Assembly
C8796826	- Container, Ammunition, Fiber, M375 for Initiators, Cartridge Actuated: Delay, M4A1, M6A1, M12A1 or M33
C8796827	- Box, Packing, Ammunition for Initiators, Cartridge Actuated: Delay, M4A1, M6A1, M12 or M33 in Fiber Container, M375
IEL-8595816	- Inspection Equipment List for Initiator, Cartridge Actuated: Delay, M4A1, Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Firing pin release.- The firing pin of the assembled initiator, less safety pin, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inch, maximum.

3.3 Initiator pin security.- Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.4 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.6 Functional.

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3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 3 feet of aircraft hose, Size No. 4, in accordance with MS28741.

TABLE I

Temperature	Pressure (PSI)	Ignition Delay (See 6.3)
-65° to -70°F	1300 to 2200	2.50 Seconds, max.
65° to 75°F	1300 to 2200	1.80 to 2.20 Seconds
160° to 165°F	1300 to 2200	1.60 Seconds, min.

3.6.2 The initiator shall not fail to fire.

3.6.3 The initiator shall function without mechanical failure (see 6.2).

3.7 Workmanship.- Requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

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4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-414, as applicable.

4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with accepted cartridges.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Table II. All non-conforming material shall be rejected. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. Four power (4X) magnification may be used for inspection of such defects as burr and foreign matter.

TABLE II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 800	50	0	1	1	2
801 - 1800	90	1	2	2	3

4.2.2.1 Cartridge, Delay, M46, Drawing D8594075, covering a detail of Drawing F8595816.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Foreign matter	Visual
MINOR	
201. Case scratched, dented, wrinkled or bulged	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Drawing F8595816, less Drawings D8593314, D8593313, D8594075 and B8359536.

CRITICAL	
1. Mixed initiators	Visual
2. Initiator pin not in cocked position	Visual
3. Safety pin missing or improperly engaged	Visual
MAJOR	
101. O-ring missing or not lubricated	Visual
102. Thread damaged	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Assembly, Drawing F8595816, less Drawings D8593314 and B8359536.

CRITICAL	
1. Cartridge loose	Manual and Visual
2. Cartridge retainer loose	Manual and Visual
3. Metering groove of cartridge retainer missing	Visual
4. Elongated hole of cartridge retainer missing	Visual

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4.2.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Drawing F8595816, less Drawing B8359536.

CRITICAL	
1. Location of cartridge retainer	Gage
2. Foreign matter in gas port	Visual
MAJOR	
101. Length from cap boss face to gas port face, max.	Gage
102. Protrusion of initiator pin, max.	Gage
103. Set screw missing, loose or above flush	Manual and Visual
MINOR	
None defined	

4.2.2.5 Assembly, Drawing F8595816.

CRITICAL	
1. Base of firing pin housing not in complete contact with face of cartridge	Manual
2. Any part missing, except as otherwise classified	Visual
3. Improper assembly	Visual
4. Marking missing, incorrect or illegible	Visual

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4.2.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Height of cap, max.	Gage
102. Initiator pin not free to rotate	Manual
103. Safety pin loose or deformed	Manual and Visual
104. Set screw not covered with barrier material	Visual
MINOR	
201. Total length, max.	Gage
202. Location of labels incorrect	Visual
203. Shipping cap missing	Visual
204. Protective finish inadequate or defective	Visual

4.2.2.6 Fiber Container, Drawing C8796826.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min.	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Packing Box, Drawing C8796827.

CRITICAL

None defined

MAJOR

None defined

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4.2.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3:

4.2.3.1 Firing pin release.- The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Initiator pin security.- The initiator pin of each subassembly shall be subjected to the initiator pin security test. Failure of any subassembly to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.- The initiator chamber and cap of each assembled initiator shall be subjected to the breakaway torque application. Failure of any assembly to comply with the requirements of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Residual magnetism.- Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.5 shall be cause for rejection of the initiator.

4.2.3.5 Functional tests.- The following tests shall be performed by the Government.

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4.2.3.5.1 Ballistics.- Twenty-one units shall be tested to determine compliance with the requirements of 3.6.1, 3.6.2 and 3.6.3. The units shall be temperature conditioned as provided in Table III and 4.3.5.1.1. The variable sampling plan of MIL-STD-414, Example C2 or C3, as applicable, for the AQL listed in Table III shall be used for each sample of 7 units. Failure of any unit to comply with the requirements of 3.6.1, 3.6.2 and 3.6.3 or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size	AQL
-65° to -70°F	7	0.65%
65° to 75°F	7	0.65%
160° to 165°F	7	0.65%

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8595816.

4.3 Test methods and procedures.

4.3.1 Firing pin release.- A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the applicable fixture which registers the amount of pull.

4.3.2 Initiator pin security.- A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the applicable fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirements specified in 3.4 shall be used in conjunction with a fixture to determine the adequacy of the joint. The initiator body shall be secured in an air vise or holding fixture. The torque shall be applied to the cap.

4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

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4.3.5 Functional.

4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.3.5.1.1.

4.3.5.1.1 Temperature conditioning.- A standard M4A1 initiator, loaded with a dummy M46 cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8796826 and C8796827.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

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Preparing activity:

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Project No. 1377-A257

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